Pygame CE: Lesson Series

Teacher overview • Scotland S3 onwards

Ten lessons of around 50-60 minutes. Learners build one small game, adding a feature each lesson. Assumes basic Python: variables, `if` statements and loops. The sequence develops programming, computational thinking, testing and creative problem solving; it is adaptable rather than tied to a particular qualification.

Setup

Install Pygame CE and import it as `pygame`:

```
pip install pygame-ce
```

```
import pygame
```

Learners create a project folder for their game. Images and sounds go in an `assets` subfolder (needed from Lesson 7). Check installation before the first lesson.

Lesson overview

#	Focus	Learners can...
1	Move a shape on screen	Understand the game loop and use keyboard input to change an object's position.
2	Smooth movement	Use the game loop to make movement continuous while a key is held.
3	Keep the player on screen	Use comparisons and min/max to enforce screen boundaries.
4	Add a target and detect collision	Create a second object and detect when two rectangles overlap.
5	Add a score	Store game information in a variable and display text in the window.
6	Add a timer	Use elapsed time to create a timed challenge.
7	Images and simple animation	Load an image and use changing values to create animation.
8	Sound and game states	Use sound and a state variable to manage different parts of a game.
9	Organise, test and debug	Make code easier to understand and test by separating repeated tasks.
10	Build and present a small game	Plan, build, test and present an independent project.

Suggested lesson routine

- **5-10 min:** recap, or predict what the code will do.
- **25-35 min:** build.
- **10-15 min:** test, share and reflect.
- Learners save a working version after each lesson (lesson1.py, lesson2.py...).

Teaching tips

- Each lesson PDF has a complete, cumulative program so learners (or you) can catch up.

- Encourage prediction before running code, and changing one thing at a time.
- Differentiate with the Support and Stretch suggestions in each lesson.
- Lesson 7 needs an image and Lesson 8 needs a .wav file: prepare these in advance.